Starlink & Ubiquiti PowerBeam Integration Guide

Technical Specifications & Networking Best Practices

Best PowerBeams to Use Alongside Starlink

When extending a Starlink connection (100 - 250+ Mbps) to another building, farm, or site, Ubiquiti PowerBeam and airMAX wireless bridges are highly reliable Point-to-Point (PtP) solutions.

1. Ubiquiti PowerBeam AC Gen 2 (PBE-5AC-Gen2 / PBE-5AC-ISO-Gen2)

Best overall for long-distance 5 GHz

- **Throughput:** Up to 450+ Mbps
- **Range:** Up to 15–25 km (Clear Line of Sight)
- **Why it fits Starlink:** Uses a 400 mm dish reflector to focus the 5 GHz radio beam, cutting through environmental noise. The ISO version adds an integrated RF isolation shield to prevent local Wi-Fi channel crowding.

2. Ubiquiti PowerBeam 60G / 60G LR (PBE-60G-LR)

Best for Multi-Gigabit speeds & zero RF noise

- **Throughput:** 1+ Gbps (Gigabit Full Duplex)
- **Range:** Up to 2 km
- **Why it fits Starlink:** Operates on the 60 GHz frequency band, bypassing congested 2.4/5 GHz spectrums. Provides near-zero latency and high bandwidth, ensuring no bottlenecks.

3. Ubiquiti NanoBeam AC Gen 2 (NBE-5AC-Gen2)

Best alternative for shorter distances (< 2 km)

- **Throughput:** Up to 450+ Mbps
- **Range:** Up to 3–5 km
- **Why it fits Starlink:** Smaller footprint, easier ball-joint mounting, and identical 450+ Mbps airMAX AC speeds for shorter property jumps where a full dish is overkill.

Key Requirements for Pairing with Starlink

- **Hardware Connection:** Starlink Gen 2 (Actuated Dish) requires the official Starlink Ethernet Adapter. Gen 3 / Starlink Mini have built-in Ethernet ports to connect directly to the PowerBeam PoE injector.
- **Bridge Configuration:** Set one PowerBeam as Access Point (PtP) and the receiver as Station (PtP) via airOS using a 40 or 80 MHz channel width on 5 GHz.
- **Downstream Networking:** Connect the receiving PowerBeam at the far end to a local Wi-Fi Access Point, switch, or router to distribute the bridged connection.

PowerBeam 5AC Gen2 (PBE-5AC-GEN2) Specifications

Key Specifications

Category	Specification
Max TCP/IP Throughput	450+ Mbps
Max Range	25+ km (15+ miles) with clear Line-of-Sight
Antenna Gain	25 dBi (400 mm / 15.7 in dish reflector)
Operating Frequency	5150 – 5875 MHz (Worldwide)
Management Radio	2412 – 2472 MHz (Dedicated 2.4 GHz Wi-Fi for UISP app)
Max TX Power	24 dBm

System & Hardware

- **Processor:** MIPS 74Kc
- **Memory:** 64 MB DDR2
- **Network Interface:** (1) 10/100/1000 Mbps Ethernet Port
- **Status LEDs:** Power, Ethernet, and (4) Signal Strength indicators

- **Operating System:** airOS 8
- **Modes:** Point-to-Point (PtP), Point-to-Multipoint (PtMP) Access Point / Station

Power & Electrical

- **Power Method:** 24V Passive PoE (Pairs 4, 5+; 7, 8 Return)
- **Power Supply:** 24V, 0.5A Gigabit PoE Adapter (Included)
- **Max Power Consumption:** 8.5W
- **Voltage Range:** 20 to 26 VDC
- **ESD/EMP Protection:** Air: ± 24 kV, Contact: ± 24 kV

Supported Channel Widths

- **PtP Mode:** 10 / 20 / 30 / 40 / 50 / 60 / 80 MHz
- **PtMP Mode:** 10 / 20 / 30 / 40 MHz

Mechanical & Environmental Specs

Property	Value
Dimensions	420 × 420 × 230 mm (16.54 × 16.54 × 9.06 inches)
Weight	2.22 kg (4.89 lbs)
Enclosure Materials	Outdoor UV-Stabilized Plastic (Feed) / Powder-Coated SPCC (Dish)
Mounting	Pole-Mounting Kit Included (Tilt adjustment: ± 20°)
Wind Loading	380 N @ 200 km/h (85.4 lbf @ 125 mph)
Wind Survivability	200 km/h (125 mph)
Operating Temperature	-40°C to 70°C (-40°F to 158°F)
Operating Humidity	5 to 95% Non-condensing
Certifications	CE, FCC, IC, RoHS